MIL-F-80233B
10 May 1985
SUPERSEDING
MIL-F-80233A
13 December 1977

MILITARY SPECIFICATION

FURNACES, VACUUM, HEAT TREATING INTEGRAL QUENCH

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers vacuum heat-treating furnaces of the electrically heated, fully automatic cycle, cold wall design, with an integral liquid and gas quench system.

1.2 Classification. The furnaces shall be of the following types. The type required shall be as specified (see 6.2.1).

Type I - Horizontal heating chamber with integral quench, in and out design.

Type II - Horizontal heating chamber with integral quench, straight through design.

Type III - Horizontal heating chamber with integral quench, straight through design, optional loading at either end.

1.2.1 Size. The size of the furnace chamber shall be as specified (see 6.2.1). The minimum chamber size shall be 6 inches wide x 12 inches in length x 6 inches high. Size changes shall increase in increments of 6 inches in any dimension, as required. The dimensions specified shall be the minimum inside effective working area of the heating chamber in which performance requirements specified herein shall be accomplished. The furnace shall be capable of processing a charge through the entire system equal in size to the chamber dimensions specified.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Industrial Plant Equipment Center, ATTN: DIPEC-SSM, Memphis, Tennessee 38114-5297, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 3424

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:32Z
Check the source to verify that this is the current version before use.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

* 2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-F-3296 - Forges, Furnaces, and Ovens (Exclusive of Space Heating and Cooking): Packaging of.

MIL-H-6875 - Heat Treatments of Steels (Aero Space Practice).

STANDARDS

FEDERAL

FED-STD-H28 - Screw Thread Standards for Federal Services.

FED-STD-376 - Preferred Metric Units for General Use by the Federal Government.

* 2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

U. S. DEPARTMENT OF LABOR

OSHA 2206 - General Industry, OSHA Safety and Health Standards (29 CFR 1910)

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402.)

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:32Z
Check the source to verify that this is the current version before use.

* 2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. The issue of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

AMERICAN NATIONAL STANDARDS INSTITUTE

ANSI/ASME	- Boiler and Pressure Vessel Code, Section VIII, Pressure Vessels, Division 1 and Division 2.
ANSI/NEMA MG 1	- Motors and Generators.
ANSI/NEMA ICS 1	- Industrial Control and Systems.
ANSI/NFPA 86D	- Industrial Furnaces, Vacuum Atmosphere.

(Application for copies should be addressed to the American National Standards Institute, ATTN: Sales Dept., 1430 Broadway, New York, NY 10018.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS

ASTM-D-3951	- Commercial Packaging, Standard Practice For

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race St., Philadelphia, PA 19103.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Design. The furnace and equipment shall be new and of the manufacturer's current design. The operating temperature range of the furnace shall be from 1000°F or less through not less than 2400°F. The furnace shall be of the vacuum type, electrically heated, of the cold wall design

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:32Z
Check the source to verify that this is the current version before use.

with a water cooled shell. The furnace shall be of the integral quench design and provide for optional liquid quenching or gas cooling. Design shall provide for programming any desired vacuum pressure and heating rate within the capability of the system specified herein. The vacuum cycle shall provide for the pumping system to evacuate the furnace chamber to pre-selected pressure before the heating cycle starts. The heating cycle shall provide for controlled power input and preheat soak periods. The liquid quench cycle shall provide for selection of quenching of the work load in either a vacuum or gas backfill condition over the oil. The design and construction of the furnace and related equipment shall conform to the requirements of NFPA Standard 86D.

* 3.1.1 Type I. The type I furnace shall be a horizontal "in and out" design having a single door located in the end of the chamber for loading and unloading in a horizontal plane. The furnace shall have not less than two chambers, the heating chamber and cooling chamber. All chambers shall be enclosed in a vacuum shell. The cooling chamber shall incorporate the cooling zone, liquid quench tank, elevator and transfer mechanism. The chambers shall be completely separated by vacuum tight door valves. Design shall provide for automatically transferring the work load through the selected heating cycle, quench or cooling cycle and positioning the load at the door for unloading.

* 3.1.2 Type II. The type II furnace shall be a horizontal "straight through" design having a door in the front for loading and a door in the rear for unloading. The furnace shall have not less than two chambers, the heating chamber and a cooling chamber. All chambers shall be enclosed in a vacuum shell. The heating chamber shall be located in the front and the cooling chamber in the rear. The cooling chamber shall incorporate the cooling zone, liquid quench tank, elevator and transfer mechanism. The chambers shall be completely separated by vacuum tight door valves. Design shall provide for automatically transferring the work load from the heating chamber through the cooling or quench cycle and positioning the load at the rear door for unloading.

3.1.3 Type III. The type III furnace shall be a horizontal straight through design incorporating three chambers enclosed in a vacuum shell. The chambers shall be individually operated and completely separated by vacuum tight door valves. Chambers may be either front loaded or side loaded through the end chambers. One end chamber shall be a gas cooling zone, the middle chamber shall be a heating zone and the other end chamber shall be a liquid quench zone. The design shall provide for two-directional flow of the work load. The system in one direction shall automatically transfer work from the loading zone (gas cooling chamber) to the heating chamber, to the oil quench chamber and to the oil quench zone for exiting. Alternative flow shall provide for loading through the oil quench zone, transferring work to the heat chamber, from the heating chamber to the gas quench zone for cooling and exiting from the gas quench chamber. Design shall allow multiple work loads to be in the furnace system at any given time.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:32Z
Check the source to verify that this is the current version before use.

* 3.1.4 Measurement systems. Unless otherwise specified, either the U.S. Customary System of Units (US) or the International System of Units (SI) shall be used in the design and construction of the furnace. When only one system of measurement is acceptable, the particular system required shall be as specified (see 6.2.1). In this specification, all measurements, dimensions, sizes and capacities are given in U.S. units. These measurements may be converted to SI units through the use of the conversion factors and methods specified in FED-STD-376.

* 3.1.4.1 Measuring and indicating device calibrations. Unless otherwise specified, either the U.S. Customary System of Units (US) or the International System of Units (SI) shall be used to graduated measuring and indicating devices such as scales, pressure gauges, temperature indicators, and other similar devices. When only one system of graduation is acceptable, the particular graduation required shall be as specified (see 6.2.1). Regardless of the measurement system used, all measuring and indicating devices on the furnace shall be graduated in the same system.

* 3.1.4.1.1 Dual calibrations. When specified (see 6.2.1), measuring and indicating devices shall be graduated in both the US and the SI System of Measurements.

3.1.5 Reclaimed materials. The furnace may contain reclaimed materials provided such materials will not jeopardize the intended use and performance. The reclaimed materials shall have been reprocessed, remanufactured or recycled in a manner which will restore them to the same chemical composition and physical properties as the materials originally selected for use on the furnace.

* 3.1.6 Energy efficiency. The furnace and all its applicable components that directly consume energy in normal operation shall be designed and constructed for the highest degree of energy efficiency as governed by the latest developments available with in the industry.

* 3.1.7 Safety and health requirements. The furnace shall be provided with safety features and devices for the protection of personnel, equipment, and the work being processed. The safety system of the furnace shall conform to NFPA Standard 86D and with the standards promulgated under OSHA 2206 applicable to the furnace itself. Additional requirements for safety and health shall be as specified (see 6.2.1 and 6.2.3).

3.1.8 Mercury restriction. The furnace shall not contain mercury or mercury compounds nor be exposed to free mercury during manufacture.

3.1.9 Asbestos restriction. Asbestos and materials containing asbestos shall not be used on or in the furnace.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:32Z
Check the source to verify that this is the current version before use.

* 3.1.10 Environmental protection. The furnace shall be designed and constructed such that, under the operating, service, transportation and storage conditions described herein, the furnace shall not emit materials hazardous to the ecological system as prescribed by Federal, state or local statutes in effect at the point of the installation.

* 3.1.11 Lubrication. Means shall be provided to insure adequate lubrication for all moving parts. Recirculating lubrication systems shall include a filter which is cleanable or replaceable. Each lubricant reservoir shall have means for determining fluid level. All oil holes, grease fittings and filler caps shall be accessible.

3.1.12 Interchangeability. To provide for replacement of worn or damaged parts, all parts shall be manufactured to definite dimensions and tolerances that will permit installation of replacement parts without modification to the part or the machine.

3.2 Construction. The furnace shall be constructed of parts which are new, without defects and free of repairs. The structure shall be capable of withstanding all forces encountered during operation of the furnace to its maximum rating and capacity without permanent distortion.

3.2.1 Castings and forgings. All castings and forgings shall be free from defects such as scale and mismatching. Processes such as welding, peening, plugging, or filling with solders or pastes shall not be used on castings or forgings to reclaim any defective components for use on the furnace. Such processes may be used only for enhancing surface finish and appearance.

3.2.2 Welding, brazing or soldering. Welding, brazing, or soldering shall be employed only where specified in the original design. None of these processes shall be employed as a repair measure for any defective part.

3.2.3 Fastening devices. All screws, pins, bolts, and other fasteners shall be installed in a manner to prevent change of tightness. Fastening devices subject to removal or adjustment shall not be swaged, peened, staked, or otherwise permanently installed.

3.2.4 Surfaces. All surfaces shall be clean and free of sand, dirt, fins, sprues, flash, scale, flux, and other harmful or extraneous materials. All edges shall be either rounded or beveled unless sharpness is required to perform a necessary function. Except as otherwise specified herein, the condition and finish of all surfaces shall be in accordance with the manufacturer's commercial practice.

3.2.5 Painting. All surfaces to be painted shall be cleaned of all foreign matter. Unless otherwise specified (see 6.2.1), the furnace shall be painted in accordance with manufacturer's standard commercial practice which shall be not less than one coat of primer and one coat of finish color.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:32Z
Check the source to verify that this is the current version before use.

3.2.6 Threads. All threaded parts used on the machine and its related attachments and accessories shall conform to FED-STD-H28 and the applicable "Detailed Standard" section referenced therein.

* 3.2.7 Dials. Dial diameters shall be such that graduations may be read from the normal operating position. Dials shall be permanently and legibly engraved or etched on a nonglare background.

3.2.8 Plates. All words on instruction and indicating plates shall be in the English language. Characters shall be engraved, etched, embossed or stamped in boldface on a contrasting background.

3.3 Electrical equipment. The furnace shall be furnished complete with electrical motors, contactors, controls, relays, transformers, single lights, and switches prewired where possible for operation upon connection to source of power. Except as otherwise specified herein, all electrical equipment and wiring shall be in accordance with NEMA ICS.

3.3.1 Power supply. Unless otherwise specified (see 6.2.1), furnaces with a power input in excess of 30 KW shall operate from a single service 460 volt, 60 Hz, 3 phase primary power supply and furnaces with power input of 30 KW and less shall operate from a single service 230 volt, 60 Hz, 3 phase primary power supply. All power actuated, manually adjusted, and automatic controls shall operate on 115 volt, 60 Hz power supply.

3.3.2 Motors. Motors rated in excess of 1/3 horsepower shall be of the dual voltage, 230/460 volt, 3 phase, 60 Hz type. All motors (except special vacuum sealed motors) shall conform to NEMA MG-1.

3.3.3 Electrical connections. Electrical connections of conductors and terminal parts shall be of the pressure or solder type. When soldered connections are used, the conductors and terminal parts shall be mechanically secured before soldering. Flux for soldering shall be of a rosin base.

3.4 Performance. Each furnace shall be capable of continuous operation with maximum loads at all vacuum pressures and operating temperatures specified for the furnace. Each furnace shall perform all vacuum pressure cycles, heating cycles, quenching and cooling cycles within the capability of the program controls and instruments specified for the furnace.

3.4.1 Load capacity. The maximum load for each furnace shall be not less than 75 lbs. per square foot of effective chamber working area at a temperature of 2000°F, and not less than 60 lbs. per square foot at 2400°F. When heavier load capacities are required, the load requirements in lbs. per square foot of hearth area shall be as specified (see 6.2.1).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:32Z
Check the source to verify that this is the current version before use.

3.4.2 Heating capacity. Each furnace shall be capable of heating 60 lbs. of iron or steel per square foot of effective chamber working area from a temperature of 75 ± 10°F to an indicated control temperature of 2000°F in not more than one hour. The load shall consist of parts or pieces having an equivalent of 1-inch or less cross section area.

3.4.3 Temperature uniformity. With the furnace charged and temperature and pressure stabilized, the temperature variation from control point to any other point within the effective chamber working area shall not vary more than ±10°F. Each furnace shall be capable of maintaining the temperature uniformity specified with any size charge up to maximum load at any desired temperature between 1200°F and 2400°F and chamber vacuum pressure within the control range of the systems.

3.4.4 Vacuum system.

3.4.4.1 Operating pressure. Unless otherwise specified the operating vacuum pressure of each furnace shall be not more than 20 microns Hg. When specified, the operating vacuum pressure of each furnace shall be as specified (see 6.2.1).

* 3.4.4.2 Pumping capacity. Empty, dry and out-gassed furnaces with work chamber volumes up to 30 cubic feet shall attain the specified operating pressure in not more than 20 minutes. Furnaces with work chamber volumes larger than 20 cubic feet shall attain the specified operating pressure in not more than 30 minutes. The vacuum chamber pressure shall be constant within +5% of operating pressure. With the furnace blanked off at minimum pressure, the heating chamber leakage rate shall not exceed 5 microns per hour.

* 3.4.4.3 Oil quench capacity. The capacity of the oil quench system shall be adequate to hold a maximum load plus one gallon of oil per pound of steel. Transfer of a charge from the heating chamber and complete immersion of the charge in the quenching oil shall be accomplished in 25 seconds or less.

3.5 Components and systems.

3.5.1 Shell. The furnace shell shall be either of one piece construction or multiple chambers joined together to from the shell. The furnace chamber incorporating the heating zone shall be of double wall construction and designed for full water jacket cooling. The cooling chamber, incorporating the cooling zones, (liquid and gas) chamber end and doors shall be water jacket cooled, tube traced or forced cooled as necessary to assure conformance with the performance requirements specified herein and to limit the temperature on all surfaces of the shell to a maximum of 150°F. The furnace chambers ends, and doors shall be designed and fabricated in accordance with Section VIII, Division 1 of ASME Boiler and Pressure Vessel Code. Unless otherwise specified (see 6.2.1), formal ASME approval and certification shall not be required. The inner shell and all structural material exposed to the vacuum environment shall be constructed of material that will not oxidize, or constructed of materials that have been plated, painted, or coated to prevent oxidization under all furnace operating temperatures, atmospheres and vacuum pressures specified herein.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:32Z
Check the source to verify that this is the current version before use.

3.5.2 Doors. Unless otherwise specified, the external doors may be either of the side hinge, top hinge, vertical lift or sliding type suspended from tram rail. When a specific type is required, the door design shall be as specified (see 6.2.1). The type III furnace shall have internal vacuum tight door valves to separate the heating chamber from the cooling and quench chamber. Each external door shall have viewing ports.

* 3.5.3 Heating chamber. The heating area shall be surrounded with either a series of radiation heat shields, multiple layers of high purity graphite felt, ceramic fiber insulation or a combination thereof. The material used shall be of a heat resistant type that will not sag, rupture, or crack and shall be designed to minimize warpage due to expansion and contraction. The chamber shall be heated by either molybdenum rod or sheet type elements or high purity graphite type elements. The insulation or radiation shields and heating elements shall be designed and arranged to minimize heat loss and assure the temperature uniformity and heating requirements specified herein. The chamber shall be provided with a hearth or workload supports constructed of molybdenum or graphite material capable of supporting maximum furnace loads without visible deflection. When specified (see 6.2.1), the chamber shall be provided with molybdenum radiation shields and heated by molybdenum rod or sheet type heating elements, with no insulating material exposed to the vacuum environments.

3.5.4 Quench system. Each furnace shall be provided with an integral mounted power operated transfer mechanism and elevator system to automatically transfer the workload through the selected quench cycle. The oil quench system shall have means of limiting the quench oil temperature increase to a maximum of 45°F, when quenching a maximum load of steel at a temperature of 2000°F. The system shall be capable of lowering the temperature of the quench oil not less than 30°F in one hour or less. Requirements of the system shall be based on lowering the temperature of a maximum charge of steel from 2000°F to not higher than 200°F with quench oil bath capacity of not less than one gallon of oil per pound of steel. The oil temperature shall be controlled and the system shall provide for selecting any desired temperature between 80°F and 180°F. Each quench system shall be provided with an oil circulating system capable of circulating the capacity of the quench system through the quench area not less than four times per minute. Design and performance of the oil quench system shall assure conformance with the severity of quench requirements specified in MIL-H-6875. The quench tank shall be provided with an indicating temperature gage and an adjustable excess temperature audible warning system with a range from 100°F to not less than 400°F, automatic temperature controls in pressure type water and oil cooling systems, quench oil level indicator and low oil level audible warning device, and an interlock to prevent start of the quenching and shut off the heat source in case of low level conditions. The excess temperature control system shall be interlocked to prevent quenching in case of excessive quench medium temperature. A limit switch shall be interlocked into the load transfer mechanism to prevent transfer of the load from the heat chamber to the quench rack unless the quench rack is in the proper position. When specified (see 6.2.1), the oil quench system shall be provided with an electrical heating system capable of raising the quench oil temperature from 75°F to 180°F in not more than 10 hours.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:32Z
Check the source to verify that this is the current version before use.

3.5.5 Water cooling system. The water cooling system shall include the apparatus and equipment used to cool the heating chamber walls, electrical terminals, seals and related equipment when operating from a water supply having a temperature of 65± 5°F. The cooling system shall be provided with the necessary controls to maintain the specified surface temperature and control condensation in the furnace chamber and vacuum systems. When specified (see 6.2.1), the water cooling system shall have a mechanical refrigeration system, or a cooling tower type cooling system as specified. If either of these systems are required, the system shall have adequate provision for cooling the water and recirculating the cooled water to the furnace and all accessory equipment. These systems, when required, shall be capable of maintaining the same surface temperature as specified for the furnace when operating from a utility water supply.

3.5.6 Gas cooling system. Each furnace shall provide for back filling the chamber with an inert gas. The cooling system shall consist of the necessary blowers, fans or gas injection nozzle system and control valves to distribute and circulate argon or other gas through the workload and chamber for uniform cooling of the workload. With the furnace operating at a temperature of 2000°F with a maximum load as specified in 3.5.2 for not less than 30 minutes, the cooling system shall cool the load from 2000°F to 3000°F in not more than 60 minutes using argon gas. The system shall provide for optional vacuum cooling, partial pressure from 20 microns Hg to 1000 microns Hg gas cooling, or gas cooling from a pressure of 380 mm Hg to not less than 500 mm Hg. The system shall provide for program control cooling of the work load in the heating chamber either under vacuum or partial pressure. The system shall operate automatically through program controls and switches with either mode selected. If fans or blowers are used, program requirements shall be as specified (see 6.2.1). When specified, a heat exchanger shall be provided to recirculate the cooling gas and expedite the cooling of the workload, and the cooling rate of the workload shall be as specified (see 6.2.1).

3.5.7 Vacuum system. Each furnace shall be provided with an automatically operated vacuum system. The system shall consist of the necessary pumps, manifolds, valves, traps, controls, gages and piping to provide the operating vacuum pressure specified for furnace chambers. The system shall have a pneumatically actuated high vacuum valve that will automatically close in case of power failure. When specified (see 6.2.1), the system shall be provided with dust traps or filters to prevent dust and particles from entering the system. If a diffusion pump is required to meet the vacuum pressure requirements specified, the pump shall be water cooled, provided with an excess temperature audible warning device and have a visual gauge for indicating oil level.

* 3.5.7.1 Vacuum ionization gauge. Furnaces designed to operate with a chamber pressure of 10 microns Hg or less shall be provided with an ionization vacuum gauge to monitor chamber pressure from not less than 10 microns to minimum pressure.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:32Z
Check the source to verify that this is the current version before use.

3.5.8 Thermocouples. Type "S" thermocouples shall be used as the sensing device for instrumentation. One thermocouple shall be used with each indicating recording and controlling instrument and one with each excess temperature control instrument. Lead wires shall be not less than 25 feet in length and of the same material as the thermocouples or of a material having the same thermoelectric characterisitics. When specified (see 6.2.1), the chamber shall have 8 type "K" thermocouples mounted through the shell for calibrating temperature uniformity in the work zone.

3.5.9 Multi-zone control. When necessary to maintain the temperature uniformity and performance requirements specified for the furnace, the chamber shall be divided into a suitable number of control zones and the required instruments and controls, conforming to the requirements specified herein, provided for multi-zone control.

* 3.5.10 Program controls. Each furnace shall be provided with instruments and controls to program, indicate, record and automatically control the complete vacuum, heating, and cooling systems performance. The vacuum control system for each furnace shall provide for selecting any desired pressure between 1000 microns Hg and 50 microns; initiate the heating cycle to any desired pressure within the control range; and provide for raising the pressure to any desired level by the introduction of inert gas at any preset temperature during the heating cycle. Unless otherwise specified, the heating cycle shall provide for one pre-heat soak period from 1 minute to not less 55 minutes, and a high heat period of not less than 6 hours with controlled heating rates for each period. Different or additional heating cycles shall be provided when specified (see 6.2.1). Each furnace system shall provide for optional gas quenching or liquid quenching of the workload. The liquid quench cycle shall provide a quench period from 1 minute to not less than 55 minutes and shall elevate the workload to the drain position at the end of selected quench time. The type II furnace, after immersion of the load into the bath, shall introduce an inert gas into the chamber to equalize the pressure and lower the chamber temperature to a preset temperature for reloading while the preceding load is being processed through the quench cycle. The type III furnace shall allow for multiple workloading in the various chambers at any time and shall provide for programming and sequencing of workloads for semi-continuous flow-through operation beginning from either end of the chamber. All cycle sequences shall be interlocked through pressure and electrical devices. In addition to the automatic system, each furnace control system shall be provided with manual over-ride controls for use in testing or emergency conditions.

* 3.5.10.1 Microprocessor. When specified (see 6.2.1), the furnace control system shall incorporate a microprocessor digital programmable/controller to provide maximum control flexibility within the limitations of the furnace's designed capability. The microprocessor shall provide for not less than 50 read/write programmable segments of random access memory and input to the system shall be selectable for either manual keyboard or magnetic tape cassette.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:32Z
Check the source to verify that this is the current version before use.

* 3.5.10.2 Instruments. Each indicating and recording instrument shall be marked to designate the particular type of thermocouple for which it was calibrated. Automatic reference junction compensation shall be provided in all instruments using thermocouples as the primary temperature sensing elements. External thermocouple lead lengths shall not affect calibration of the instruments other than for minor adjustments. Instruments shall be designed for mounting on the panels so that any tendencies toward parallax in reading will be alleviated. When specified (see 6.2.1), all instruments shall be of the digital readout design and both the vacuum and temperature recorders shall be of the printout design.

* 3.5.10.3 Temperature controller. The temperature control system shall be either of the saturable core reactor, variable reactance transformer, or silicon controlled rectifier type. The control system shall limit power input to any desired maximum and shall be accurate within 1/2 of one percent of full scale reading and capable of maintaining the set control temperature within a maximum deviation of $\pm 2^{\circ}F$.

3.5.10.3.1 Temperature control indicator. The control indicator scale width shall be not less than the recorder chart width. The scale range shall be not less than 100 degrees F above the maximum operating temperature of the furnace. The increments between successive temperature marks shall be not greater than 20 degrees F with the spacing of numbered marks not greater than 400 degrees F. The control indicator shall be provided with two pointers. One pointer shall be adjustable to the desired control temperature, and one pointer shall indicate the actual furnace temperature.

3.5.10.4 Temperature recorder. The temperature recorder shall be of the strip chart, self-balancing type and shall be accurate within 0.3 percent of full scale reading.

3.5.10.5 Vacuum controller and recorder. Each furnace shall be provided with a strip chart vacuum controlling-recording system to control and record heating chamber operating pressure. The chart shall be calibrated in microns Hg with maximum readability from 1 microns to 1000 microns. The control indicator shall be provided with 2 pointers. One pointer shall be adjustable to the desired pressure, and one pointer shall indicate the actual vacuum pressure. The vacuum control system shall provide safety provisions that will affect power shut-off of the furnace heating elements should vessels pressure exceed the preset pressure.

3.5.10.6 Charts. The width of the chart for the vacuum and temperature recorder shall be not less than 9 inches, and the length shall be not less than 80 feet to the roll. The standard chart speed shall be 1 inch per hour with provisions for changing speed to four and six inches per hour. The pen mechanism shall drive the pen across the entire chart width in not more than 30 seconds. A platen shall be provided behind the chart plane to permit the marking of notation on the chart while the instrument is in operation. The chart mechanism shall be furnished in a dust proof housing with a hinged door. The door shall be provided with a glass that will permit full view of the chart width and not less than 7 inches of chart length.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:32Z
Check the source to verify that this is the current version before use.

3.5.10.7 Excess temperature control. The furnace shall be provided with an excess temperature controller. The excess temperature control circuit shall be independent of the control circuity. The controller shall provide for selecting the over-temperature control point, and shall function to shut off the heat source to the furnace when the over-temperature control point is reached. In case the furnace is shut down by the excess temperature control system, the system shall sound an audible alarm, and the furnace shall remain off until started manually. Calibrated accuracy shall be within one percent of full scale reading.

* 3.5.10.8 Control panel. Each furnace shall be provided with a floor mounting, factory wired, control panel conforming to NEMA ICS[1] for mounting all instruments and accessory controls. Two NEMA-1 fusible safety switches or circuit breakers shall be provided in the control system, one for the main incoming power to the heating elements and one for the pumps, motors, and control system. Signal lights shall be provided to indicate the location and completion of the load through the furnace operation. The panel shall provide for either semi-flush mounting or flush mounting of all instruments. When specified (see 6.2.1), the control panel shall be provided with shock absorbing mounts.

3.6 Work trays. When required, work trays shall be provided. The trays shall be constructed of material that will not vaporize under the vacuum and temperature of the furnace, shall provide maximum circulation of quench oil through the work, and shall be capable of handling maximum furnace loads. Tray dimensions and number required shall be as specified (see 6.2.1).

3.7 Nameplate. Unless otherwise specified (see 6.2.1), a corrosion resistant metal nameplate shall be securely attached to each furnace. The nameplate shall contain the information listed below. If the furnace is a special model, the model designation shall include the model of the basic standard furnace and a suffix keyed to the manufacturer's permanent records.

Nomenclature
Manufacturer's name
Manufacturer's model designation
Manufacturer's serial number
Power input characteristics and ratings
Operating vacuum pressure range (microns Hg)
Effective chamber dimensions
Quench tank capacity (gallons)
Operating temperature range (degrees F)
Date of manufacture
Contract or order number
National Stock Number or Plant Equipment Code (NSN Preferred)
U S

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:32Z
Check the source to verify that this is the current version before use.

3.8 Technical data. Technical data shall be furnished as specified (see 6.4).

3.9 Workmanship. Workmanship of the furnace and accessories shall be of a quality equal to that of the manufacturer's commercial equipment of the type specified herein.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Quality conformance inspection (see 4.3).
b. Acceptance test (see 4.4).

4.3 Quality the conformance inspection. Quality conformance inspection shall be applied to the furnace prior to being offered for acceptance under the contract. Unless otherwise specified (see 6.2.1), quality conformance inspection shall consist of the examination in 4.5, the tests in 4.6.1, and 4.6.2 and the inspection in 4.7. Failure of the furnace to pass the examination, test, or inspection shall be cause for rejection.

* 4.4 Acceptance test. Acceptance test shall be performed on the furnace to insure conformance with this specification. Unless otherwise specified (see 6.2.1), the acceptance test shall be performed only after the furnace is installed at its final location, and the test shall consist of the examination in 4.5 and all test in 4.6. Failure of the furnace to pass one or more tests or any part of a test shall be cause for rejection.

* 4.5 Examination. The furnace and equipment shall be examined for design, conformance to NFPA Standard 86D, dimensions, construction, materials, components, electrical equipment, and workmanship to determine compliance with the requirements of this specification.

* 4.6 Tests. All tests shall be conducted at an ambient temperature of 77 degrees F ±18 degrees F. Test instruments and sensing elements used to perform the tests specified for the furnaces and instruments shall have been calibrated with a standard potentiometer type instrument and a calibrated thermocouple or sensing devices of known accuracy within the previous six months to assure an accuracy within ±2 degrees F.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:32Z
Check the source to verify that this is the current version before use.

4.6.1 Instrument test. Indicating, recording and control instruments shall be tested by impressing various predetermine voltages, simulating thermocouple outputs for different furnace temperatures to terminals where normally the thermocouples would be connected. The instruments shall be tested through their operating range. Each circuit shall be tested to determine compliance with 3.5.10.2 through 3.5.10.7.

4.6.2 Circuit test. Each circuit of the electrical system shall be tested for continuity, short circuits, and dielectric strength in accordance with NEMA ICS-1 as applicable.

* 4.6.3 Operational test. The furnace shall be prepared for operation and operated in accordance with the manufacturer's handbook of instructions. After the system has dried and been out-gassed, the furnace shall be cooled to a chamber temperature of 77 ± 18°F. The regulating controls shall be set to attain minimum chamber pressure and maximum temperature. The vacuum system shall evacuate the furnace chamber to the selected pressure in a time limit not greater than the time limit specified for the size furnace under test. With the furnace operating at maximum rated temperature, the outside surface temperature shall be checked at random locations, and shall not exceed 150°F. The furnace shall be operated at minimum chamber pressure and all systems shall be turned off. The leak-up rate test shall be applied to each heating chamber with the adjoining chambers at atmospheric pressure. The leak-up rate shall not exceed 5 microns per hour.

* 4.6.4 Performance test. The furnace shall be loaded with a charge of iron or steel equivalent to 60 lbs per sq foot of hearth area. The controls shall be programmed for the furnace to operate at minimum chamber pressure and maximum power input with the heating cycle to start at a chamber pressure of 200 microns Hg. At the end of one hour from the start of the heating cycle, the work chamber temperature shall be 2000°F (±10). After the furnace has cooled to ambient temperature, with a chamber load of 60 lbs per sq. ft. of hearth area, the controls shall be programmed for the furnace to perform the following cycle sequence:

a. Pump down to minimum pressure
b. Begin temperature cycle at 500 microns Hg and continue pump down to minimum operating pressure
c. Soak for 12 minutes at 1800°F at minimum pressure
d. After 12 minutes, increase the temperature to 2000°F
e. At a temperature of 1850°F, vacuum pressure raised to 200 microns HG
f. Soak for 30 minutes at 2000°F with a chamber pressure of 200 microns Hg
g. Transfer load to quench chamber and forced gas cooling cycle using argon gas

Source: https://assist.dla.mil -- Downloaded: 202640-26T04:32Z
Check the source to verify that this is the currentversion before use.

The system shall cool the workload from the 2000°F to 3000°F in one hour or less. The furnace shall operate automatically through the specified cycle sequence and record the complete performance. At least 3 additional performance tests shall be run to a temperature of 2000°F, with the furnace fully charged and different individual cycles selected for each performance test. The heating elements and safety interlocks shall be blocked or by-passed and a maximum load processed through a liquid quench cycle to assure the quench system conforms with the requirements specified herein. Failure of the furnace to meet any requirement specified herein or any evidence of damage to the furnace or equipment shall be cause for rejection. When specified (see 6.2.1), the quench tank shall be filled with the proper oil and a load processed through the quench system and the system checked for compliance with 3.5.4.

*4.6.5 Temperature uniformity test. The furnace shall be loaded as specified in 4.6.4 and a chamber temperature survey performed to determine conformance with the uniformity requirements specified for the furnace. A minimum of nine test locations shall be used with eight thermocouples located symmetrically through the charge so as to scan the work space. One thermocouple shall be located within six inches of each control thermocouple. The survey shall be performed with equilibrium established at 1200°F operating temperature and at 2000°F. Three survey readings shall be taken at intervals of five to ten minutes under different chamber pressure in the range between 1000 microns Hg and minimum chamber pressure. The maximum temperature variation between the instrument control point in the chamber and all other points under test shall vary not more than plus or minus 10°F. Type "K" thermocouples may be used to perform the above survey.

4.7 Inspection of packaging. Packaging shall be inspected to determine compliance to the requirements of section 5.

5. PACKAGING

5.1 Packaging, packing and marking. Unless otherwise specified, packaging, packing, and marking shall be in accordance with ASTM-D-3951. When specified preservation, packing and marking shall be in accordance with specification MIL-F-3296. Level of preservation and packing shall be as specified (see 6.2.1).

6. NOTES

6.1 Intended use. Furnaces covered by this specification are intended for such heat treating operations as hardening, tempering, annealing, normalizing, brazing, degassing and sintering in vacuum atmosphere with optional liquid or gas quenching.

6.2 Ordering data.

6.2.1 Procurement requirements. Purchasers should specify their requirements in procurement documents, including whether each choice is required or not required, by entering an appropriate statement identified to each of the following:

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:32Z
Check the source to verify that this is the current version before use.

a. Title, number and date of this specification.

b. Type furnace required (see 1.2).

c. Size required (specify heating chamber length, width, and height (see 1.2.1).

d. If furnace is required to be configured in a specific measurement system (US or SI), state required system (see 3.1.4).

e. If measuring and indicating devices are required to be graduated in a specific measurement system (US or SI), state required system (see 3.1.4.1).

f. Dual calibrations (US and SI), if required (see 3.1.4.1.1).

g. Additional safety and health requirements, if required (see 3.1.7).

h. Painting, if different (see 3.2.5).

i. Electrical system and input power supply if different (see 3.3.1).

j. Maximum load, if different specify (see 3.4.1).

k. Operating vacuum pressure, if different specify microns Hg (see 3.4.4.1).

l. ASME Certification of Compliance of Boiler and Pressure Vessel Code, when required (see 3.5.1).

m. Door design, if different (see 3.5.2).

n. Radiation shields and molybdenum heating elements only, when required (see 3.5.3).

o. Oil quench system electric heaters, when required (see 3.5.4).

p. If mechanical refrigeration system is required, so specify (see 3.5.5).

q. If cooling tower system is required, so specify (see 3.5.5).

r. Program requirements, if different cooling modes required (see 3.5.6).

s. Heat exchanger and cooling rate, when required (see 3.5.6).

t. Vacuum system dust taps of filters, when required (see 3.5.7).

u. Test thermocouples, when required (see 3.5.8).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:32Z
Check the source to verify that this is the current version before use.

v. Heating cycle program requirements, if different (see 3.5.10).

w. Microprocessor controller, if required (see 3.5.10.1).

x. Digital readout instruments, if required (see 3.5.10.2).

y. Digital printout recorders, if required (see 3.5.10.2).

z. Shock absorbing of instruments panel, when required (see 3.5.10.8).

aa. Trays and fixtures, when required (see 3.6).

bb. Nameplate, if different (see 3.7).

cc. Quality conformance inspection if different (see 4.3).

dd. Acceptance test if different (see 4.4.)

ee. Oil quench load test, if required (see 4.6.4).

ff. Packaging and packing, if different (see 5.1).

6.2.2 Contract data requirements. Required technical data such as, operator's manuals, parts lists, wiring diagrams, foundation and anchor bolt plans and acceptance test reports should be specified on a DD Form 1423, Contract Data Requirements List, incorporated into the contract.

6.2.3 Safety and health requirements. Paragraph 3.1.7 requires compliance only with those OSHA requirements that concern the furnace itself. It does not require compliance with those OSHA requirements that concern "the furnace in its operating environment" such as noise levels, radiation levels electromagnetic emissions, noxious vapors, air contaminants, and heat. Since OSHA limits the total hazard level of these hazards in the environment (and does not limit the hazard level of individual furnace in the environment) the procuring activity is advised to analyze the existing hazard levels in the proposed operating environment, and specify additional requirements that will integrate the new furnace into its future operating environment. The above, and any other additional safety and health requirements should be specified in detail under 6.2.1.g.

6.2.4 Marginal notations. The margins of this specification are marked with asterisks to indicate where changes (addition, modifications, corrections, deletions) from previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractor are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:32Z
Check the source to verify that this is the current version before use.

Custodians:
Army - AL
Navy - SH
Air Force - 99

Review activities:
Army - AL
Air Force - 99
DLA - GS

User activities:
Navy - OS, AS

Preparing activity:
DLA - IP

Project Number:
3424-0104

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:32Z
Check the source to verify that this is the current version before use.

INSTRUCTIONS: In a continuing effort to make our standardization documents better, the DoD provides this form for use in submitting comments and suggestions for improvements. All users of military standardization documents are invited to provide suggestions. This form may be detached, folded along the lines indicated, taped along the loose edge *(DO NOT STAPLE)*, and mailed. In block 5, be as specific as possible about particular problem areas such as wording which required interpretation, was too rigid, restrictive, loose, ambiguous, or was incompatible, and give proposed wording changes which would alleviate the problems. Enter in block 6 any remarks not related to a specific paragraph of the document. If block 7 is filled out, an acknowledgement will be mailed to you within 30 days to let you know that your comments were received and are being considered.

NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

(Fold along this line)

(Fold along this line)

DEFENSE LOGISTICS AGENCY
Headquarters
Defense Industrial Plant Equipment Center
Memphis, Tennessee 38114-5297

NO POSTAGE NECESSARY IF MAILED IN THE UNITED STATES

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

BUSINESS REPLY MAIL
FIRST CLASS PERMIT NO. 4966 WASHINGTON D. C.
POSTAGE WILL BE PAID BY THE DEFENSE LOGISTICS AGENCY

Commander
Defense Industrial Plant Equipment Center
ATTN: DIPEC-SSM
Memphis, Tennessee 38114-5297

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:32Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-F-80233B	Furnaces, Vacuum, Heat Treating Integral Quench

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426

PREVIOUS EDITION IS OBSOLETE.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:32Z
Check the source to verify that this is the current version before use.